



WHITEPAPER

# Sovereign cloud: architecture patterns

In-country, air-gapped and in-region AI — how sovereignty becomes a property of the architecture

**In-country. Air-gapped. In-region AI.**

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Deployment classes, segregation under central policy, offline updates, FIPS-validated cryptography, WORM-backed audit and Whale AI inside the perimeter — the patterns behind the Government Edition.

**Sovereign Cloud** is the deployment for estates that must prove where data lives and who can reach it: the control plane runs in-country, on your own cloud, on-premises or fully air-gapped, with an offline update channel and no call-home. Whale AI runs on in-region models inside the perimeter, and the audit trail is WORM-backed, so sovereignty is a property of the architecture rather than a clause in a contract.

| Air-gapped                                                                       | In-region AI                                         | 4 + in-country                                                   |
|----------------------------------------------------------------------------------|------------------------------------------------------|------------------------------------------------------------------|
| Deployment class with signed offline update bundles and no outbound connectivity | Whale AI Abyss on local models — zero external calls | SaaS regions (SIN · BOM · FRA · LAX), or BYOC / on-prem anywhere |

## The problem regulators pose

Data-residency law and national regulators demand to know exactly where data lives, who operates it, and who has access — which most hyperscaler-only deployments cannot prove.

## The pattern, end to end

### 1 Pattern 1

The platform is deployed inside national borders — BYOC, on-premises or fully air-gapped.

### 2 Pattern 2

An air-gapped control plane runs with no outbound connectivity; updates arrive as signed offline bundles.

- 3

**Pattern 3**  
Residency rules are enforced on every resource and record; directorates are segregated under central policy.
- 4

**Pattern 4**  
In-region models deliver Whale AI without data leaving the jurisdiction.
- 5

**Pattern 5**  
Sovereignty is evidenced on demand from the WORM-backed audit trail and residency controls.

## Sovereign Operations — the module behind it

|                                                                                                                                                                                               |                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Segregation under central policy</b><br>Ministries, directorates or agencies operate their own estates with autonomy inside national guardrails — one policy plane, many isolated domains. | <b>Air-gapped classes</b><br>The most sensitive estates run fully disconnected with an offline update channel; hybrid classes connect to sovereign clouds such as AWS GovCloud, Azure Government and Google Distributed Cloud. |
| <b>Hardened by default</b><br>FIPS-validated cryptography, MFA mandatory on every role, PAM with session recording always on, residency enforced and evidenced.                               | <b>Offline-tolerant edge</b><br>Remote sites keep operating under headquarters-grade policy when disconnected and reconcile when they reconnect.                                                                               |

## How it is operated

- 1

**Classify estates**  
Assign each estate a deployment class — in-country connected, sovereign cloud, or air-gapped.
- 2

**Segregate and govern**  
Directorates get isolated domains; policy, identity and audit remain central.
- 3

**Evidence continuously**  
State-audit evidence and compliance packs (FedRAMP / IRAP / StateRAMP-style) are generated from the WORM-backed trail.

## What national regimes demand, and the control that answers

| Regime                      | What it demands                                                    | Control                                                                                         |
|-----------------------------|--------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|
| National data-residency law | Data, inventory and audit records stay inside the border, provably | In-country deployment classes; residency enforced per resource and evidenced from the audit log |
|                             |                                                                    |                                                                                                 |

| Regime                                                           | What it demands                                          | Control                                                                        |
|------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Security accreditation (FedRAMP / IRAP / StateRAMP-style)</b> | Control evidence assembled to an accreditation framework | Compliance pack export from the WORM-backed audit trail; FIPS-validated crypto |
| <b>Air-gap mandates for classified estates</b>                   | No outbound connectivity, controlled update path         | Air-gapped deployment class with signed offline update bundles                 |
| <b>Privileged-access rules</b>                                   | Every administrative session attributable and reviewable | PAM always-on with session recording; MFA mandatory on every role              |

## The Government Edition sovereignty layer

- Full Enterprise platform, sovereign/air-gapped deployment
- PAM always-on, MFA mandatory, FIPS-validated crypto
- Immutable audit log (WORM), data residency enforcement
- Compliance pack export (FedRAMP/IRAP/StateRAMP-style accreditation evidence)
- Sovereign cloud connectors: AWS GovCloud, Azure Gov, GDC, national clouds

## Sovereign hyperscaler regions

AWS GovCloud, Azure Government, Google Distributed Cloud and national sovereign clouds connect through the same connectors and are governed by the same policy plane as private and air-gapped estates. Whale AI runs on in-region models with no external calls; on the Government Edition it runs fully offline.

## Questions from accreditation and procurement teams

### 01 How do updates reach an air-gapped estate?

Signed release bundles are transferred through your approved process and verified on import. Nothing calls home, and there are no forced upgrades on sovereign estates.

### 02 Does Whale AI work without internet?

Yes. The Government Edition runs Whale AI Abyss on in-region models inside the perimeter, with no external calls.

### 03 Can we connect sovereign hyperscaler regions?

Yes — AWS GovCloud, Azure Government, Google Distributed Cloud and national sovereign clouds connect through the same connectors, governed by the same policy plane.

### 04 How is it procured and priced?

Tender- and empanelment-ready on 3–5 year fixed-bid terms, priced as the Enterprise base plus the sovereignty layer, directly or through accredited implementation partners.

## 05 How do updates reach an air-gapped deployment?

As signed release bundles transferred through your approved process and verified on import. There is no call-home and no forced upgrade on sovereign estates.

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## 06 Which accreditations does it support?

Compliance pack export produces evidence for FedRAMP-, IRAP- and StateRAMP-style accreditation; the platform itself holds ISO/IEC 27001, 27017, 27018, 27701 and ISO 22301 certifications.

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